



Mathematics Part 1 — Cbse Periodic Test (Pt) — 40 Marks

Subject: Mathematics Part 1 | Class: Class 9

Total Marks: 40 | Time Allowed: 1h 30m | CBSE

General Instructions:

1. Read all questions carefully before answering.
2. All questions are compulsory unless stated otherwise.
3. Marks for each question are indicated on the right.
4. Write neatly and legibly.

Q.NO.	QUESTIONS	MARKS
Section A Attempt all 8 questions. Each carries 1 mark(s).		
1	What is the distance between the points (3, 5) and (7, 5)? (A) 2 (B) 12 (C) 4 (D) $\sqrt{32}$ <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	1
2	The ancient Egyptian formula for the area of a circle was $A \approx (8d/9)^2$, where d is the diameter. This can also be expressed as: (A) $(64/81)r^2$ (B) $(128/81)r^2$ (C) $(256/81)r^2$ (D) $(32/81)r^2$ <i>[Ch 6: Measuring Space: Perimeter and Area]</i>	1
3	The belief that if a random event happens multiple times in a row, the opposite outcome is more likely to happen next, is known as: (A) Law of Large Numbers (B) Gambler's Fallacy (C) Subjective Probability (D) Sampling Error <i>[Ch 7: The Mathematics of Maybe: Introduction to Probability]</i>	1



Q.NO.	QUESTIONS	MARKS
4	When we change the values of b while keeping the value of a fixed in the equation $y = ax + b$, what happens to the lines? (A) They intersect at the origin (B) They shift but remain parallel to the original line (C) They become perpendicular (D) They have different slopes <i>[Ch 2: Introduction to Linear Polynomials]</i>	1
5	Assertion (A): Linear decay describes a pattern in which a quantity decreases by a fixed amount over equal intervals. Reason (R): In a linear relationship represented by $y = ax + b$, the constant b represents the slope, and a straight line representing linear decay has a positive slope. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 2: Introduction to Linear Polynomials]</i>	1
6	If one pushes a chord away from the centre, at some stage the chord becomes a single point. What is the distance of the centre from this point? (A) Zero (B) Equal to the chord length (C) Half the radius (D) The radius <i>[Ch 5: I'm Up and Down, and Round and Round]</i>	1
7	Since $x^4 - y^4 = (x^2 - y^2)(x^2 + y^2)$, which of the following must be a factor of $x^4 - y^4$? (A) $x + y$ only (B) $x^2 + y^2$ only (C) $x - y$ (D) $x^2 - y^2$ only <i>[Ch 4: Exploring Algebraic Identities]</i>	1
8	Assertion (A): The point $(3, -5)$ lies in Quadrant II. Reason (R): Points in Quadrant II have a negative x-coordinate and a positive y-coordinate. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	1



Q.NO.	QUESTIONS	MARKS
Section B Attempt all 4 questions. Each carries 2 mark(s).		
9	The cost of a journey is given by $C(d) = 100 + 60d$. By what constant amount does the cost increase per km, and what does this represent? <i>[Ch 2: Introduction to Linear Polynomials]</i>	2
10	In a survey of 50 students, 15 students said they liked football. The number of students who like football is 15, and the ____ (frequency/relative frequency) is ____ (fill in the fraction or decimal). <i>[Ch 7: The Mathematics of Maybe: Introduction to Probability]</i>	2
11	In which quadrant does a point with a negative x-coordinate and a positive y-coordinate lie? <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	2
12	State Brahmagupta's rule regarding the product of two debts and express it in modern notation. <i>[Ch 3: The World of Numbers]</i>	2
Section C Attempt 3 out of 4.		
13	The village playground is shaped as a square of side 40 metres. A path of width s metres is created around the playground for people to walk. Find an expression for the area of the path in terms of s. <i>[Ch 4: Exploring Algebraic Identities]</i>	3
	OR Use the Baudhāyana–Pythagoras theorem to show why Theorem 6 must be true. <i>[Ch 5: I'm Up and Down, and Round and Round]</i>	
14	Find the 10th term of the GP: 5, 25, 125, <i>[Ch 8: Predicting What Comes Next: Exploring Sequences and Progressions]</i>	3
15	What value of π was provided by Āryabhaṭa in 499 CE, and what profound insight did he express about this value using the word 'asanna'? Also, write Mādhava's exact formula for π. <i>[Ch 6: Measuring Space: Perimeter and Area]</i>	3
Section D Attempt 2 out of 3.		
16	A gym charges a fixed monthly fee and an additional cost per hour for using the badminton court. When the court is used for 10 hours, the bill is ₹800, and when it is used for 15 hours, the bill is ₹1100. The monthly bill y depends on the hours of use x by the relation $y = ax + b$. Find the values of a and b. Identify the slope and the y-intercept of the line, and explain what they represent in this context.	5



Q.NO.	QUESTIONS	MARKS
	<i>[Ch 2: Introduction to Linear Polynomials]</i>	
	OR Brahmagupta introduced the concepts of "fortunes" (Dhana) and "debts" (Riña) to represent positive and negative numbers. Using these real-world concepts, explain why: (i) the product of two debts is a fortune, and (ii) subtracting a negative number is the same as adding a positive number. Provide a numerical example for each. <i>[Ch 3: The World of Numbers]</i>	
17	A city has two main roads which cross each other at the centre of the city. These two roads are along the North-South (N-S) direction and East-West (E-W) direction. All the other streets of the city run parallel to these roads and are 200 m apart. There are 10 streets in each direction. (i) Using 1 cm = 200 m, draw a model of the city in your notebook. Represent the roads/streets by single lines. (ii) There are street intersections in the model. Each street intersection is formed by two streets—one running in the N-S direction and another in the E-W direction. Each street intersection is referred to in the following manner: If the second street running in the N-S direction and 5th street in the E-W direction meet at some crossing, then we call this street intersection (2, 5). Using this convention, find: (a) how many street intersections can be referred to as (4, 3). (b) how many street intersections can be referred to as (3, 4). <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	5
Section E Attempt 1 out of 1.		
18	Case Study: Room Layout with Coordinates Reiaan is planning his room layout on a coordinate plane. The room door extends from D₁ (9.5, 0) to R₁ (11.5, 0). The bathroom door extends from B₁ (0, 1.5) to B₂ (0, 4). A rectangular study table has three of its feet at the points (8, 9), (11, 9) and (11, 7). Reiaan wants to verify the dimensions of each item and check whether the bathroom door is wider or narrower than the room door. He also needs to find the position of the fourth foot of the table to complete the layout. (i) What is the width of the room door? (ii) What are the coordinates of the fourth foot of the rectangular study table? (iii) Find the width of the bathroom door. Is the bathroom door narrower or wider than the room door? (iv) Both B₁ and B₂ have x-coordinate equal to 0. What does this indicate about the position of the bathroom door with respect to the y-axis? <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	5

